

10.1 Testing Methodology

A comprehensive multi-layered testing approach was adopted to ensure the reliability, security, and usability of the AgriTrace platform. The testing process encompassed the following methodologies:

1. **Unit Testing:** Individual functions and components were tested in isolation to verify correct behavior. Firebase service functions (e.g., `createListing`, `claimListing`, `calculateCarbonOffset`) were tested with various input combinations.
2. **Integration Testing:** End-to-end workflows were tested to ensure proper interaction between modules. For example, the complete listing lifecycle (creation → claiming → pickup → delivery → recycling) was tested across all three user roles.
3. **Static Analysis:** ESLint with TypeScript plugin performed automated code quality checks. The `npm run lint` command was integrated into the development workflow to catch issues early.
4. **Type Checking:** TypeScript's strict mode (`npm run typecheck`) verified type safety across all 94 source files, catching potential runtime errors at compile time.
5. **Security Testing:** Firestore security rules were tested using the Firebase Emulator Suite. Unauthorized access attempts were simulated for all collections to ensure proper role-based access control.
6. **Cross-Browser Testing:** The application was tested on Chrome 120+, Firefox 121+, Safari 17+, and Edge 120+ to ensure rendering consistency and feature compatibility.
7. **Responsive Testing:** All pages were verified across desktop (1920×1080), tablet (768×1024), and mobile (375×667) viewports using Chrome DevTools device emulation.
8. **Performance Testing:** Page load times, Firestore query latency, and image upload speeds were benchmarked to ensure acceptable performance under typical usage conditions.

10.2 Test Cases

Comprehensive testing was performed across all modules of the AgriTrace platform. The following table presents key test cases:

ID	Module	Test Case	Expected Result	Actual Result	Status
TC01	Authentication	Valid email/password login	Redirect to dashboard	Redirected to dashboard	Pass